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EDUCATION

Sardar Vallabhbhai National Institute of Technology **Surat, Gujarat, 2024 – 2026**
Master of Technology,Computer Science and Engineering, CGPA:8.57
Specialization: Information Security and Privacy

Netaji Subhas University of Technology **New Delhi, 2020 – 2024**
Bachelor of Technology, Information Technology , CGPA: 7.59
Specialisation: Network and Information Security

TECHNICAL SKILLS

Languages: C, C++, Python, HTML, CSS, Javascript, Typescript, QT
Frameworks/Libraries: ReactJS, NextJS, ThreeJS, NodeJS, ExpressJS, Framer Motion, TailWindCSS, Shadcn
Database/Tools: MySQL, PostgreSQL, MongoDB, Firebase, Supabase, Clerk, Resend, Edgestore, VScode, Pycharm
Coursework: Object Oriented Programming, Computer Networks, Operating System, DBMS

PROFESSIONAL EXPERIENCE

Siemens Digital Industries Software | *C++, NX, Unit Testing* **Jul 2026 – Present**
Software Engineer – Associate

- Engineered **license decomposition for NX Assembly Constraints**, enabling feature-level licensing and reducing unnecessary dependency on higher-tier Assembly capabilities.
- Designed and implemented **cross-layer license management** covering checkout, release, and stateful licensing across commands, undo operations, part closure, and session lifecycle.
- Extended and validated **Assembly and Design Center workflows**, ensuring consistent licensing behavior across diverse feature execution paths and application states.

Siemens Digital Industries Software | *C++, NX, Unit Testing* **May 2025 – Jun 2026**
Student Intern

- Engineered an **AI-powered Assembly Search** framework supporting structured queries, multi-condition filtering, result aggregation, and efficient retrieval across complex product structures.
- Expanded Assembly Search with **rich component properties and metadata**, including physical properties, spatial information, component states, and assembly-specific attributes.
- Developed **advanced Assembly and positioning functionality**, improving feature workflows while implementing and validating licensing behavior for assembly-related capabilities.
- Integrated and enhanced **assembly data retrieval and presentation**, including descriptive component information, search result enrichment, and contextual tooltips.
- Implemented **performance instrumentation and automated testing** for assembly workflows, while debugging data-flow, runtime, and integration issues across multiple software components.

IIT Kanpur | *Python, Regex, Small Language Models, LLM, Reinforcement Learning* **May 2025 – Sep 2025**
Research Intern

- Built a web scraping pipeline using Python to collect legal templates and clauses from online sources for AI tasks.
- Designed an automated timekeeping system for lawyers that uses regex to classify and log case activities; prototyping a small language model to generate and refine regex patterns for more accurate categorization.
- Developing a legal AI framework to generate templates and retrieve clauses based on user input, enabling dynamic document creation and adaptation.
- Implementing an LLM-based template modification module with reinforcement learning from user feedback to reduce hallucinations and enforce legal compliance.

SalesForce | *LWC, API, Apex Testing, CLI Setup, Process Automation* | Certificate **Oct 2023 - Nov 2023**
Developer Intern

- Played a pivotal role in the establishment of a **Salesforce Org**, driving organizational efficiency.
- Developed intricate validation rules and workflow automation using Salesforce's **flow/process builder**.

PERSONAL PROJECTS

RetroGroove | *TailWind, ReactJs, NextJs, TypeScript, Shadcn, Supabase, PostgreSQL* | GitHub | Live

* Engineered a music platform, empowering users to **upload**, **like**, and **listen** to songs seamlessly.
* Implemented robust user authentication mechanisms using **SupaBase** and databases using **PostgreSQL**.
* Introduction of a system-wide dark mode utilizing advanced **Shadcn** and **next-themes** methodologies.

Listing | *NextJs, Prisma, TailWind, TypeScript, PostgreSQL* | GitHub | Live

* Developed an innovative hotel booking application, facilitating users to **reserve accommodations** or **list properties**.
* Introduced **advanced filtering mechanisms**, enabling users to **search** based on **location** or preferred destination.
* Utilised **multi factor user authentication** using **Next Auth** and managed database using **Prisma ORM** and **PostgreSQL**.

MediSense-AI | *LLaMA, Python, Pinecone, Streamlit, LangChain, HuggingFace* | GitHub

* Built an end-to-end **LLM-based RAG pipeline** for the medical domain using fine-tuned **LLaMA models**, Wikipedia scraping, PDF ingestion, and topic modeling.
* Integrated **Pinecone vector DB** and embedding generation with **HuggingFace** models for semantic retrieval and question-answering in a **Streamlit** chatbot UI.
* Engineered modular utilities for **text preprocessing**, document embeddings, and inference APIs, enabling scalable and domain-specific language understanding.

RESEARCH PAPERS

Eye Controlled Wheelchair | *OpenCV, Dlib, PyQt5, Imutils, Scikit-image, Scikit-learn* | GitHub

* Developed an **eye-controlled wheelchair system** using advanced eye-tracking technology to enhance **navigation for users** with **motor impairments**.
* Evaluated **machine learning models (KNN, SVM, Random Forest, Decision Tree)** for **accuracy, precision, recall, and F1 score** to **optimize performance**.
* Developed a **front-end interface using Qt**, enabling users to capture their **own eye images** for creating **custom datasets**.

Heart-Attack-Prediction-Analysis | *PyCaret, Pandas, NumPy, Matplotlib, Scikit-Learn, Google Colab* | GitHub

* Built a **machine learning pipeline** to predict heart attack risk using clinical data, automated with **PyCaret**.
* Performed **EDA and feature analysis** to identify key health indicators impacting prediction outcomes.
* Benchmarked multiple models and achieved high accuracy using Logistic Regression and Extra Trees.

Cross-Domain Fake News Detection | *PyTorch, BERT, RoBERTa, DistilBERT, Transformers, Scikit-learn* | GitHub

* Developed a **transformer-driven fake news detection framework** by fine-tuning **DistilBERT, BERT, and RoBERTa** and combining predictions through voting, probability averaging, validation-weighted fusion, threshold tuning, and stacked meta-classification.
* Improved **LIAR binary classification** from **71.03% to 80.42%** and achieved a best verified **45.91% accuracy** on six-class truthfulness classification using transformer-based ensemble strategies.
* Designed a **hierarchical binary-to-multi-class classification** approach and explored **metadata graph modelling** using speaker, subject, party, and context information while controlling for potential metadata leakage.
* Validated cross-domain and multilingual robustness on **Politifact, GossipCop, and PolyglotFakeFacts**, achieving **91.95%, 87.62%, and 99.61% accuracy**, respectively, using primary non-diagnostic methods.

Reinforcement Learning-Based Quantum Circuit Routing Optimization | *Python, RL, GNN, Transformer*

* Developed a **reinforcement learning-based quantum circuit routing framework** to minimize **SWAP-gate overhead** under hardware connectivity constraints.
* Implemented and compared **GNN and Transformer-based neural architectures** for learning routing policies using REINFORCE, action masking, reward shaping, and curriculum-based training.
* Built a **reproducible benchmarking framework** across **139 quantum circuits**, comparing neural and heuristic routing methods; the GNN model achieved **2.3× faster training** than the Transformer.

ACHIEVEMENTS

- **Google Summer of Code**: Integrated LLM and vector-based retrieval pipelines into Rocket.Chat, **enabling knowledge-grounded conversations and optimizing query understanding** through efficient indexing and embeddings.
- Hacker Cup'23 By Meta, Secured **3412 Global rank** in Round 2.
- CodeForces **Max Rating-1903, Candidate Master**, Ranked 167th in Round 954, 307th in Epic Round (Div 1+2).
- LeetCode **Max Rating-2329, Guardian**, Ranked 298th in Biweekly 117, 272th in Weekly 373.
- CodeChef **Max Rating-2025, 5 Star**, Ranked 59th in Starter 86, 11th in Starter 100, 51st in Starter 117.
- Code For Freedom'23 By Coding Ninjas, Achieved **AIR 131** out of **4000+** competitors.
- CodeKaze Sep'23 By Coding Ninjas, Secured **11th College Rank** and an **AIR of 344** among **1 lakh** competitors.
- Code Gladiators By Tech Gig, Secured **59 Global rank** among **20000** competitors.
- CodeKaze Jun'23 By Coding Ninjas, Secured **19th College Rank** and an **AIR of 711** among **2 lakh** competitors.
- GeeksForGeeks Coding Score-1200, Institute Rank-131.
- HackerRank 5 Star in C++, Problem Solving.